Project 1

What needs to be done for Project 1?

1) Planning
   a) Hardware
   b) Software
2) Implementation
   a) Hardware
   b) Software
3) Test & Debug
4) Demonstrate and Deliver the report.

Planning - Hardware

What hardware you need:

- STM32 evaluation board
- Breadboard
- 19 X green LEDs for cars
- 3 (green,amber,red) X LEDs for traffic light
- 21 Resistors for LEDs
- Potentiometer
- 1 Resistor for potentiometer
- 3 X Shift Register (74HC164)
- Wires for connecting elements

Planning- Software

What functions you need to write?

1. Initializes the traffic light GPIO pins
2. Initializes the GPIO pins that are connected to Serial to parallel shift registers that drive LEDs representing cars.
3. Initializes the ADC peripheral that the Potentiometer that adjusts the traffic load is connected to
4. Clear Lane (reset shift registers)
5. Update Traffic light e.g., update_traffic(color)
6. Configure the system ready to run the demo.
   a. The clock configuration can be done here if it was not done before main() was called.
7. Create the queue used by the queue send and queue receive tasks.
   a. xQueueCreate()
   b. Refer to https://www.freertos.org/a00116.html for more details.
8. Add to the registry, for the benefit of kernel aware debugging.
   a. vQueueAddToRegistry ()
   b. Refer to https://www.freertos.org/vQueueAddToRegistry.html for more details.
9. Start the tasks and timer running.
   a. vTaskStartScheduler ()
   b. Refer to https://www.freertos.org/a00132.html for more details.